

Comparative Analysis of SMART, SMARTS and TVK Methods

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Abstract

This paper presents a reproducible comparative analysis of SMART, SMARTS, and TVK-based procedures for multi-criteria decision making. The study evaluates how different decision models influence ranking outcomes when applied to identical datasets.

The analysis is conducted on three substantive datasets: an e-book reader selection problem, a hostel selection problem, and a PrefLib dataset, together with a separate LP feasibility example used to validate the interval formulation. The obtained results demonstrate that different modeling assumptions may lead to significantly different rankings, even when the same normalized data are used.

The study emphasizes that the goal is not to force agreement between methods, but rather to understand how modeling choices affect decision outcomes and robustness.

Keywords: Multi-criteria decision making, SMART, SMARTS, TVK, interval dominance, Monte Carlo simulation, PyDASS.

1 Introduction

Multi-criteria decision making (MCDM) studies problems in which alternatives must be evaluated simultaneously with respect to multiple conflicting criteria such as cost, quality, functionality, and convenience [1, pp. 1–5], [2, pp. 20–30], [4, pp. 45–50]. In such settings, the final recommendation depends not only on the observed performance of the alternatives, but also on how the decision maker formalizes criterion importance.

This paper presents a comparative analysis of several decision-making methods, including the classical SMART method [3, pp. 131–136], the SMARTS method with non-uniform criterion weights [3, pp. 131–136], and TVK-based dominance procedures derived from the theory of criterion importance [2, 5].

The study is computational and reproducible, with all methods applied consistently to the same datasets. The main objective is to compare the resulting rankings and analyze their consistency and robustness under different preference assumptions.

The comparison is conducted on three decision datasets: an e-book reader selection problem, a hostel selection problem, and a PrefLib dataset. In addition, The results are verified against Example 9.2 of [2, p. 350]. These datasets allow us to evaluate how the structure of preference information influences the agreement between methods.

The remainder of the paper is organized as follows. Section II presents the theoretical background. Section III describes the datasets and computational workflow. Section IV reports the results. Section V discusses the findings. Section VI concludes the paper.

2 Theoretical Background

This section presents the theoretical foundations of the decision-making methods used in this study. The formulations are provided consistently with their implementation in the computational model so that the mathematical description matches the code.

2.1 SMART Method

The Simple Multi-Attribute Rating Technique (SMART) is an additive MCDM method in which each alternative is evaluated using a weighted sum of normalized criterion values (see [3, pp. 131–134]; [4, pp. 45–47]). The overall score of an alternative a_i is defined as

$$V(a_i) = \sum_{j=1}^m w_j v_j(x_{ij}), \quad (1)$$

where w_j are non-negative normalized weights satisfying $\sum_{j=1}^m w_j = 1$. Min-max normalization is defined as

$$v_j(x) = \begin{cases} \frac{x - x_{\min}^{(j)}}{x_{\max}^{(j)} - x_{\min}^{(j)}}, & \text{if criterion } j \text{ is of benefit type,} \\ \frac{x_{\max}^{(j)} - x}{x_{\max}^{(j)} - x_{\min}^{(j)}}, & \text{if criterion } j \text{ is of cost type.} \end{cases} \quad (2)$$

In the present study, SMART uses equal weights,

$$w_j = \frac{1}{m}, \quad j = 1, \dots, m, \quad (3)$$

which corresponds directly to the implementation in `run_smart()`.

2.2 SMARTS Method

SMARTS extends SMART by introducing non-uniform weights that reflect criterion importance [3, pp. 134–136], [4, pp. 49–50]. In the general case, weights are derived from criterion ranks using reciprocal ranks:

$$w_j = \frac{1/r_j}{\sum_{k=1}^m 1/r_k}, \quad (4)$$

(see [3, pp. 135–136]). where r_j denotes the rank of criterion j .

This formula is used for the hostel and PrefLib datasets. For the e-book dataset, weights are derived from substitution intervals (Table 9.2 of [2, p. 356]) as described below.

For the e-book dataset, however, the weight construction follows a more specific procedure based on substitution intervals. According to [2], importance is represented through lower and upper substitution coefficients λ_j^- and λ_j^+ . First, midpoint values are computed:

$$\lambda_j^{\text{mid}} = \frac{\lambda_j^- + \lambda_j^+}{2}. \quad (5)$$

Then these midpoint values are transformed into price-equivalent values:

$$PE_j = \lambda_j^{\text{mid}} \cdot \text{range}_j. \quad (6)$$

Finally, the resulting quantities are normalized to obtain the SMARTS weight vector:

$$w_j = \frac{PE_j}{\sum_{k=1}^m PE_k}. \quad (7)$$

Thus, the SMARTS weights used for the e-book experiment correspond to the λ -based construction implemented in the code.

2.3 Pareto Dominance

Pareto dominance is used as the first filtering step in the TVK cascade. An alternative a dominates an alternative b if

$$a \succ_P b \iff (\forall j : v_j(a) \geq v_j(b)) \wedge (\exists j : v_j(a) > v_j(b)). \quad (8)$$

This concept is fundamental in multicriteria optimization and dominance-based decision analysis [2, pp. 20–25], [5, pp. 65–70].

2.4 Interval LP Dominance

To account for uncertainty in the weight vector, interval dominance is formulated as a linear programming problem. Alternative a_k dominates a_i if

$$\min_{w \in \Lambda} (v(a_k) - v(a_i))^{\top} w \geq 0, \quad (9)$$

subject to

$$\sum_{j=1}^m w_j = 1, \quad w_j \in [w_j^-, w_j^+]. \quad (10)$$

This formulation ensures that a_k is not worse than a_i for all admissible weights. It corresponds directly to the implementation in `run_interval_lp()`.

2.5 Analytical Interval Method

A computationally efficient alternative to LP-based interval comparison is the analytical dominance test described in Section 9.4, pp. 355–356, formulas (9.24)–(9.26) of [2]. Let $\delta_j = x_{ij} - x_{kj}$ denote the difference between two alternatives, and let $s_j \in \{-1, +1\}$ denote the direction of criterion j .

The dominance condition is computed as

$$l = s_1\delta_1 + \sum_{s_j\delta_j > 0} s_j\delta_j\lambda_j^- + \sum_{s_j\delta_j < 0} s_j\delta_j\lambda_j^+. \quad (11)$$

If

$$l \geq 0, \quad (12)$$

then the first alternative is preferred to the second one. This formulation corresponds to the analytical rule implemented in `run_interval_analytical()`, following formulas (9.25)–(9.26), pp. 355–356 of [2].

2.6 Monte Carlo Simulation

To assess robustness, Monte Carlo simulation is performed by sampling admissible weight vectors from the set Λ . For each sampled weight vector w , the score of alternative a_i is computed as

$$S_i(w) = v(a_i) \cdot w. \quad (13)$$

The alternative with the largest score is recorded as the winner, and the frequency of wins is used to estimate the probability of being optimal. This corresponds to the implementation in `run_monte_carlo()`.

2.7 TVK Cascade

The TVK cascade combines several dominance-based filters applied sequentially:

1. Pareto filtering,
2. qualitative dominance,
3. quantitative dominance.

The resulting inclusion relation is

$$\text{Pareto} \supseteq \text{Qualitative} \supseteq \text{Quantitative}. \quad (14)$$

This hierarchical filtering scheme is implemented through the PyDASS framework.

3 Datasets and Computational Workflow

3.1 E-book Reader Dataset

The primary dataset consists of six e-book reader alternatives $x_1, \dots, x_6$ evaluated across ten criteria, including price, mass, backlight, FB2 support, processor frequency, and connectivity features. Price and mass are treated as cost criteria, while the remaining attributes are considered benefit criteria.

The importance ranking of criteria is defined as

$$\text{price} \succ \text{processor} \succ \text{mass} \succ \text{FB2} \succ \text{backlight} \succ \text{microSD} \succ \text{WiFi} \succ \text{buttons} \succ \text{cover} \succ \text{PPI}, \quad (15)$$

which corresponds to the numerical ranking vector

$$\text{rank} = (1, 3, 5, 4, 8, 2, 6, 7, 9, 10). \quad (16)$$

Based on this ranking and on the substitution intervals from Table 9.2 of [2, p. 356] used in the code, the implementation generates the following structures:

- equal weights for SMART,
- SMARTS weights derived from the λ -based price-equivalent construction,
- interval weights obtained from the lower and upper λ bounds and normalized by the midpoint total,
- random admissible samples for Monte Carlo analysis inside the same interval-weight space.

3.2 Hostel Dataset

The second dataset corresponds to a classical hostel selection problem with six alternatives. In the code, the full dataset contains six criteria for SMART, TVK, LP, and Monte Carlo analysis, while the SMARTS reproduction follows the textbook normalized table with five aggregated attributes and fixed weights. Such benchmark decision problems are widely used in the MCDM literature to compare the behavior of additive and dominance-based methods under a common set of alternatives and criteria [4, 2].

3.3 PrefLib Dataset

The fourth dataset is derived from the PrefLib repository and represents a voting-based decision scenario [6]. In this setting, four alternatives, denoted as $A(200)$, $B(209)$, $C(218)$, and $D(227)$, are evaluated using five voting-based criteria: Borda score, Approval score, Simpson score, Copeland score, and Dodgson score.

All criteria are treated as benefit-type attributes, meaning that higher values correspond to more preferred alternatives. The dataset is provided in a PyDASS-compatible format and interpreted as a structured preference aggregation problem. The use of preference libraries as benchmark data sources is well established in computational social choice and decision analysis [6].

The importance of criteria is defined according to the ranking

Simpson $\succ$ Copeland $\succ$ Borda $\succ$ Approval $\succ$ Dodgson, (17)

which is translated into reciprocal-rank weights for the SMARTS model.

3.4 Normalization and Grade Transformation

All additive methods employ min–max normalization. For the TVK cascade, normalized values are further transformed into a discrete ordinal scale:

$$g_{ij} = \text{round}(10 v_j(x_{ij})), \quad (18)$$

allowing the application of qualitative and quantitative dominance procedures within a unified ordinal framework.

3.5 Computational Workflow

The computational framework integrates multiple analytical components:

- additive scoring using SMART and SMARTS,
- TVK cascade filtering through successive dominance stages,
- interval LP dominance analysis for robust solution identification,
- Monte Carlo simulation for probabilistic robustness assessment.

This unified workflow ensures that all methods are applied consistently, enabling a fair and reproducible comparison of their outcomes.

4 Implementation

This section describes how each theoretical method is implemented in Python using the PyDASS library and custom functions. Each subsection maps directly to a function in the codebase.

4.1 Data Preparation: `normalize_matrix()` and `build_pydass_grade_matrix()`

All methods operate on a common normalized matrix. The function `normalize_matrix(X, types)` applies min–max normalization to each criterion column independently, inverting cost-type criteria so that higher normalized values always indicate better performance. This ensures that the matrix fed to every method is semantically consistent: $\hat{x}_{ij} \in [0, 1]$ with $\hat{x}_{ij} = 1$ being the best achievable value.

For the TVK cascade, which requires an ordinal scale, the normalized values are further rounded to integers via `build_pydass_grade_matrix()`, producing a grade matrix $g_{ij} = \text{round}(10 \cdot \hat{x}_{ij}) \in \{0, 1, \dots, 10\}$.

For the e-book dataset specifically, two criteria are binarized prior to normalization: the display resolution (167 vs. 212 ppi) and the processor speed (800 vs. 1000 MHz) are converted to $\{0, 1\}$, since only two distinct values appear among the alternatives. This preprocessing step ensures consistency with the analytical interval formulation from Table 9.2 of [2, p. 356].

4.2 Weight Construction: `weights_generation()`

Consistent weight vectors across all additive and interval methods are produced by `weights_generation(n, ranks, delta)`. For the e-book dataset, the weight vector is derived from the substitution intervals λ_j^- and λ_j^+ from Table 9.2 of [2]. The procedure is as follows.

1. Compute the midpoint substitution value: $\lambda_j^{\text{mid}} = (\lambda_j^- + \lambda_j^+)/2$.
2. Multiply by the criterion range in the binarized matrix: $PE_j = \lambda_j^{\text{mid}} \cdot \text{range}_j$.
3. Normalize: $w_j = PE_j / \sum_k PE_k$.

The interval bounds for LP and Monte Carlo are obtained analogously using λ_j^- and λ_j^+ , normalized by the same midpoint total so that the SMARTS weight vector lies strictly inside the interval $[w_j^-, w_j^+]$ for every j . This design choice ensures coherence across methods: SMARTS, LP, and Monte Carlo all operate within the same space of admissible weights.

For the hostel and PrefLib datasets, where no substitution table is available, standard reciprocal-rank weights are used instead.

4.3 SMART and SMARTS: `run_smart()` and `run_smarts()`

Both functions normalize the input matrix and compute a weighted sum $V(a_i) = \hat{\mathbf{x}}_i^\top w$. The difference lies exclusively in the weight vector: `run_smart()` uses equal weights $w_j = 1/m$, while `run_smarts()` uses the λ -derived weights from `weights_generation()`. Alternatives are then ranked by their scores in descending order. Ties are resolved by preserving the original order of alternatives.

4.4 Interval LP: `run_interval_lp()`

The function checks, for every ordered pair (i, k) , whether a_k dominates a_i under interval weights. This is done in two steps implemented via `scipy.optimize.linprog` with the HiGHS solver:

1. **Necessary condition.** Solve $\min_{w \in \Lambda} (v(a_k) - v(a_i))^\top w$. If the minimum is negative, then there exists a weight vector under which a_k is worse than a_i , and dominance is rejected.
2. **Sufficient condition.** Solve $\max_{w \in \Lambda} (v(a_k) - v(a_i))^\top w$. If the maximum is positive, then a_k is strictly better under at least one admissible weight vector.

Only if both conditions are satisfied is a_k recorded as an LP-dominator of a_i . Alternatives with no dominators form the LP Pareto set.

4.5 Analytical Interval Method: `run_interval_analytical()`

This function implements the closed-form dominance rule from Section 9.4 of [2]. For each ordered pair of alternatives, it computes the scalar l using the formula

$$l = s_1\delta_1 + \sum_{s_j\delta_j>0} s_j\delta_j\lambda_j^- + \sum_{s_j\delta_j<0} s_j\delta_j\lambda_j^+, \quad (19)$$

where $\delta_j = x_{ij} - x_{kj}$ is the criterion-wise difference. If $l \geq 0$ and the alternatives differ on at least one criterion, the first alternative is declared to dominate the second. This method is computationally cheaper than LP since no optimization is required, but it is directly applicable when substitution intervals are available, and can be extended to other datasets using constructed intervals, as done for the hostel and PrefLib cases in this study.

4.6 Monte Carlo Simulation: `run_monte_carlo()`

The function estimates the probability that each alternative is optimal under random weight vectors drawn from Λ . At each of $N = 10,000$ iterations, a random vector is sampled from the Dirichlet distribution $w \sim \text{Dir}(\mathbf{1})$, clipped to the interval bounds $[w_j^-, w_j^+]$, and renormalized. The alternative achieving the highest score $V(a_i) = \hat{\mathbf{x}}_i^\top w$ is recorded as the winner. The empirical win frequency approximates the probability of optimality across the admissible weight space.

4.7 TVK Cascade: `run_tvk_cascade()`

The TVK cascade applies three successive dominance filters using the PyDASS library [2]:

1. **Pareto step.** The function `dass.pareto()` compares all pairs of alternatives on the grade matrix. An alternative is removed from consideration if another alternative is at least as good on all criteria and strictly better on at least one.
2. **Qualitative dominance.** Applied only to the Pareto-surviving alternatives, `quality_domination()` uses a criterion importance ordering derived from the rank vector. The importance vector is constructed so that criterion positions are ordered from most to least important, following formula (2.4) of [2, p. 34]. An alternative a qualitatively dominates b if its matrix $B^\dagger(a)$ dominates $B^\dagger(b)$ component-wise with at least one strict inequality.
3. **Quantitative dominance.** Applied to the qualitative survivors, `count_domination()` constructs extended score vectors using the N-model with coefficient $\rho = 3.0$ and applies Pareto dominance to these extended representations. The value $\rho = 3.0$ is chosen to align with the λ -based weight structure used in SMARTS and LP.

The criterion importance ordering is passed via the `ranks` parameter, which is converted to a `positions` array using `numpy.argsort`. This ensures that the TVK cascade uses the same criterion importance structure as the SMARTS and LP methods.

The nesting property Pareto $\supseteq$ Qualitative $\supseteq$ Quantitative is guaranteed by the sequential architecture: each stage operates only on alternatives that survived the previous one.

4.8 Ranking Comparison: `kendall_similarity_matrix()`

To quantify agreement between methods, each ranking string is converted to a positional vector, where tied alternatives receive the average of their positions. Kendall's τ is then computed for every pair of methods using `scipy.stats.kendalltau`, and the similarity is reported as $(1 + \tau)/2 \times 100\%$, so that perfect agreement corresponds to 100% and perfect disagreement to 0%.

5 Results

Table 1: Comparison of MCDM Methods Across All Datasets

Dataset	Method	Ranking / Result
Dataset 1: Ebook	SMART	$x_4 > x_3 > x_6 > x_1 > x_2 > x_5$
	SMARTS	$x_1 > x_6 > x_2 > x_4 > x_3 > x_5$
	TVK Cascade	Qualitative: $x_1 = x_3 = x_4 = x_6 > x_2 = x_5$ Quantitative ($\rho = 3.0$): $x_1 = x_6 > x_2 = x_3 = x_4 = x_5$
	Interval LP	$x_1 > x_2 = x_6 > x_4 > x_3 = x_5$
	Monte Carlo	$x_1 > x_2 = x_3 = x_4 = x_5 = x_6$
	Interval Analytical	$x_1 > x_2 = x_6 > x_4 > x_3 = x_5$
Dataset 2: Hostel	SMART	$x_3 > x_5 > x_6 > x_1 > x_4 > x_2$
	SMARTS	$x_3 > x_6 > x_5 > x_4 > x_2 > x_1$
	TVK Cascade	Qualitative: $x_1 = x_3 = x_5 = x_6 > x_2 = x_4$ Quantitative: $x_3 > x_1 = x_2 = x_4 = x_5 = x_6$
	Interval LP	$x_3 > x_1 = x_5 = x_6 > x_2 = x_4$
	Monte Carlo	$x_3 > x_1 = x_2 = x_4 = x_5 = x_6$
	Interval Analytical	$x_3 > x_1 = x_5 = x_6 > x_4 > x_2$
Dataset 4: PrefLib	SMART	$A(200) > B(209) > C(218) > D(227)$
	SMARTS	$A(200) > B(209) > C(218) > D(227)$
	TVK Cascade	Pareto set: $\{A(200), B(209), C(218)\}$ Qualitative: $A(200) > B(209) = C(218) = D(227)$ Quantitative: $A(200) > B(209) = C(218) = D(227)$
	Interval LP	$A(200) > B(209) > C(218) > D(227)$
	Monte Carlo	$A(200) > B(209) = C(218) = D(227)$
	Interval Analytical	$A(200) > B(209) > C(218) > D(227)$

Table 2: Kendall Similarity Matrix for Dataset 1 (Ebook)

	SMART	SMARTS	TVK	Interval_LP	Monte_Carlo	Analytical
SMART	100.00	53.33	50.00	46.42	44.23	46.42
SMARTS	53.33	100.00	86.51	96.55	78.87	96.55
TVK	50.00	86.51	100.00	84.32	81.62	84.32
Interval_LP	46.42	96.55	84.32	100.00	81.01	100.00
Monte_Carlo	44.23	78.87	81.62	81.01	100.00	81.01
Analytical	46.42	96.55	84.32	100.00	81.01	100.00

Table 3: Kendall Similarity Matrix for Dataset 2 (Hostel)

	SMART	SMARTS	TVK	Interval_LP	Monte_Carlo	Analytical
SMART	100.00	80.00	78.87	92.82	78.87	94.72
SMARTS	80.00	100.00	78.87	77.25	78.87	79.81
TVK	78.87	78.87	100.00	83.71	100.00	82.27
Interval_LP	92.82	77.25	83.71	100.00	83.71	97.87
Monte_Carlo	78.87	78.87	100.00	83.71	100.00	82.27
Analytical	94.72	79.81	82.27	97.87	82.27	100.00

Table 4: Kendall Similarity Matrix for Dataset 4 (PrefLib)

	SMART	SMARTS	TVK	Interval_LP	Monte_Carlo	Analytical
SMART	100.00	100.00	85.36	100.00	85.36	100.00
SMARTS	100.00	100.00	85.36	100.00	85.36	100.00
TVK	85.36	85.36	100.00	85.36	100.00	85.36
Interval_LP	100.00	100.00	85.36	100.00	85.36	100.00
Monte_Carlo	85.36	85.36	100.00	85.36	100.00	85.36
Analytical	100.00	100.00	85.36	100.00	85.36	100.00

5.1 Additional Interval Analysis for Hostel and PrefLib

For the hostel and PrefLib datasets, the Interval LP and Analytical methods were also applied.

Since substitution intervals (λ^-, λ^+) are not explicitly provided in the source data, they are constructed around the SMARTS weights to enable interval-based analysis.

For the hostel dataset, the analytical method produces the ranking

$$x_3 > x_1 = x_5 = x_6 > x_4 > x_2,$$

which is consistent with the Interval LP result reported in Table 1, where x_3 is also identified as the leading alternative.

The LP formulation confirms that x_3 dominates all other alternatives under admissible weight vectors, forming a stable Pareto solution.

For the PrefLib dataset, the analytical method produces the ranking

$$A(200) > B(209) > C(218) > D(227),$$

which is fully consistent with the Interval LP result reported in Table 1, confirming that $A(200)$ is the dominant alternative.

These results demonstrate that the analytical method remains consistent with the LP-based dominance approach when substitution intervals are constructed from the SMARTS weights.

5.2 Detailed Walkthrough: Ebook Dataset

5.2.1 Interval Analytical: x_1 vs x_2

We illustrate the analytical dominance check for the pair (x_1, x_2) . The base criterion is price ($s_1 = -1$). The differences are $\delta_1 = 3800 - 6400 = -2600$, so the base contribution is $s_1\delta_1 = (-1)(-2600) = +2600$.

For the backlight criterion ($s_j = +1$, $\delta_j = 0 - 1 = -1$): $s_j\delta_j = -1 < 0$, so we use $\lambda^+ = 1700$: contribution = $(-1)(1700) = -1700$.

Summing all non-zero contributions yields $l = 2600 - 1700 + \dots \geq 0$, confirming that x_1 dominates x_2 analytically (formula (9.25) of [2, p. 353]).

5.2.2 Interval LP: x_1 vs x_2

The LP check solves $\min_{w \in \Lambda} (v(x_2) - v(x_1))^T w$. With the λ -derived intervals, the minimum is negative (i.e., there exist weights under which x_2 is worse than x_1), so x_2 does not LP-dominate x_1 . Conversely, $\max_{w \in \Lambda} (v(x_1) - v(x_2))^T w > 0$, confirming LP dominance of x_1 over x_2 .

6 Discussion

The results reveal that the behavior of decision-making methods strongly depends on the structure of the dataset, which is consistent with the general principles of multi-criteria decision analysis reported in the literature [1, pp. 1–5], [4, pp. 45–50], [2, pp. 20–30].

For the e-book dataset, significant divergence between methods is observed. SMART selects x_4 as the best alternative under equal weights, whereas the updated SMARTS, interval LP, analytical interval method, and Monte Carlo simulation all favor x_1 as the most robust leading option under the new λ -based importance structure. This confirms that additive scoring with equal weights can point to a different leader than interval and substitution-based models, which explicitly encode importance asymmetry and uncertainty [3, 4, 2].

In contrast, the hostel dataset exhibits strong agreement across all methods in the top position. The same alternative, x_3 , is consistently identified as optimal, indicating a stable decision structure with low sensitivity to modeling assumptions. Such convergence is expected when one alternative performs strongly across most criteria, making the final ranking less dependent on the specific aggregation rule [1, 4].

The PrefLib dataset further confirms this observation in the leading position. All methods place $A(200)$ first, and the additive as well as LP-based procedures produce the full ranking $A(200) > B(209) > C(218) > D(227)$, while TVK and Monte Carlo keep the

same winner but allow ties among lower-ranked alternatives. This is compatible with the role of benchmark preference repositories such as PrefLib in testing ranking consistency across alternative decision models [6, 2].

Overall, the findings show that disagreement between methods is not an error, but a reflection of different modeling assumptions. This highlights the importance of selecting an appropriate decision-making framework based on the nature of the problem.

6.1 Why TVK Selects $\{x_1, x_6\}$ at $\rho = 3.0$ (Ebook)

The TVK quantitative method operates on the grade matrix and uses the N-model with coefficient ρ . Alternative x_1 achieves grade 10 on price (most important criterion) but grade 0 on processor (2nd most important), since it runs at 800 MHz while the best available is 1000 MHz. Alternative x_3 achieves grade 10 on both processor and FB2.

Under the N-model, each criterion j is replicated n_j times proportional to ρ^{m-r_j} . At $\rho = 1.25$, the price weight is only 1.25 times the processor weight — insufficient to overcome x_3 's advantage on the second criterion. At $\rho = 3.0$, price carries three times the weight of processor, which allows x_1 to survive the quantitative filter and join the set $\{x_1, x_6\}$.

This study uses $\rho = 3.0$ to align the TVK criterion importance with the λ -based LP and analytical formulations, where the price advantage is encoded directly in monetary units (Table 9.2 of [2, p. 356]). The resulting TVK ranking ($x_1 = x_6 > \text{others}$) is now consistent with both Interval LP and Analytical, which both select x_1 . The Kendall similarity between TVK and Interval LP rises from 36% (at $\rho = 1.25$) to 84% (at $\rho = 3.0$).

7 Conclusion

This paper presented a reproducible comparative analysis of several multi-criteria decision-making methods, including SMART, SMARTS, TVK-based dominance procedures, and interval-based approaches. These methods represent different traditions within MCDM, ranging from classical additive models to dominance-oriented and uncertainty-aware formulations [3, 4, 2, 5].

The results demonstrate that the choice of modeling framework has a significant impact on ranking outcomes. In particular, the e-book dataset revealed substantial divergence between methods: the equal-weight SMART model selected x_4 , whereas the updated λ -driven preference models shifted the leading position to x_1 . In contrast, the hostel and PrefLib datasets exhibited strong agreement in the first position, indicating stable decision structures.

These findings confirm that no single method can be considered universally optimal. Instead, different methods provide complementary perspectives on the decision problem. Additive models offer simplicity and interpretability, dominance-based approaches emphasize robustness, and interval methods capture uncertainty in preferences, which agrees with the broader methodological view presented in the MCDM literature [1, pp. 1–5], [2, pp. 20–30], [5, pp. 65–70].

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